A120607 concise calculus certificate

VERIFIED

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Definition and first terms

$37\,A(x) = 36 + 81\,x + A(x)^{10}$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(3)*T(x); T-polynomial coefficients are the Delta multiplicities below.`

$a(0), \ldots, a(7) = 1, 3, 15, 270, 5505, 124818, 3028200, 76896180.$

Kernel used throughout: $\rho(u) = u\,(1 - 5\,u^1 - 40\,u^2 - 210\,u^3 - 756\,u^4 - 1890\,u^5 - 3240\,u^6 - 3645\,u^7 - 2430\,u^8 - 729\,u^9)$, so $x = \rho(T(x))$.

Typogeometry

1x3 {1,1}x15 {1,{1,1}}x75 {1,1,1}x120

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_{10}: 729, Δ_2: 5, Δ_3: 40, Δ_4: 210, Δ_5: 756, Δ_6: 1890, Δ_7: 3240, Δ_8: 3645, Δ_9: 2430.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_9) \in \mathbb{Z}^9_{\geq 0} : \sum_{j=1}^{9} j k_j = n - 1\}, \quad K = \sum_{j=1}^{9} k_j,$$

$$a(n) = \frac{3}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1, \ldots, k_9} 5^{k_1} 40^{k_2} 210^{k_3} 756^{k_4} 1890^{k_5} 3240^{k_6} 3645^{k_7} 2430^{k_8} 729^{k_9}.$$

$$a(n) = \frac{3}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{9} P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-9 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 9. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n,u) = N(n,u)/\rho(u)^8$, $\deg_u N = 81$.

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u))$.

Matrix dimensions: 20×20; remainder 9×10, rank 9, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.